

①

Taylor's Theorem

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2}f''(x_0)(x-x_0)^2$$

Most Important for MSCF

$$+ \dots + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n$$

$$+ \frac{1}{(n+1)!} f^{(n+1)}(c)(x-x_0)^{n+1}$$

c is between
 x_0, x

Error or
Remainder

Special Case $n=0$

②

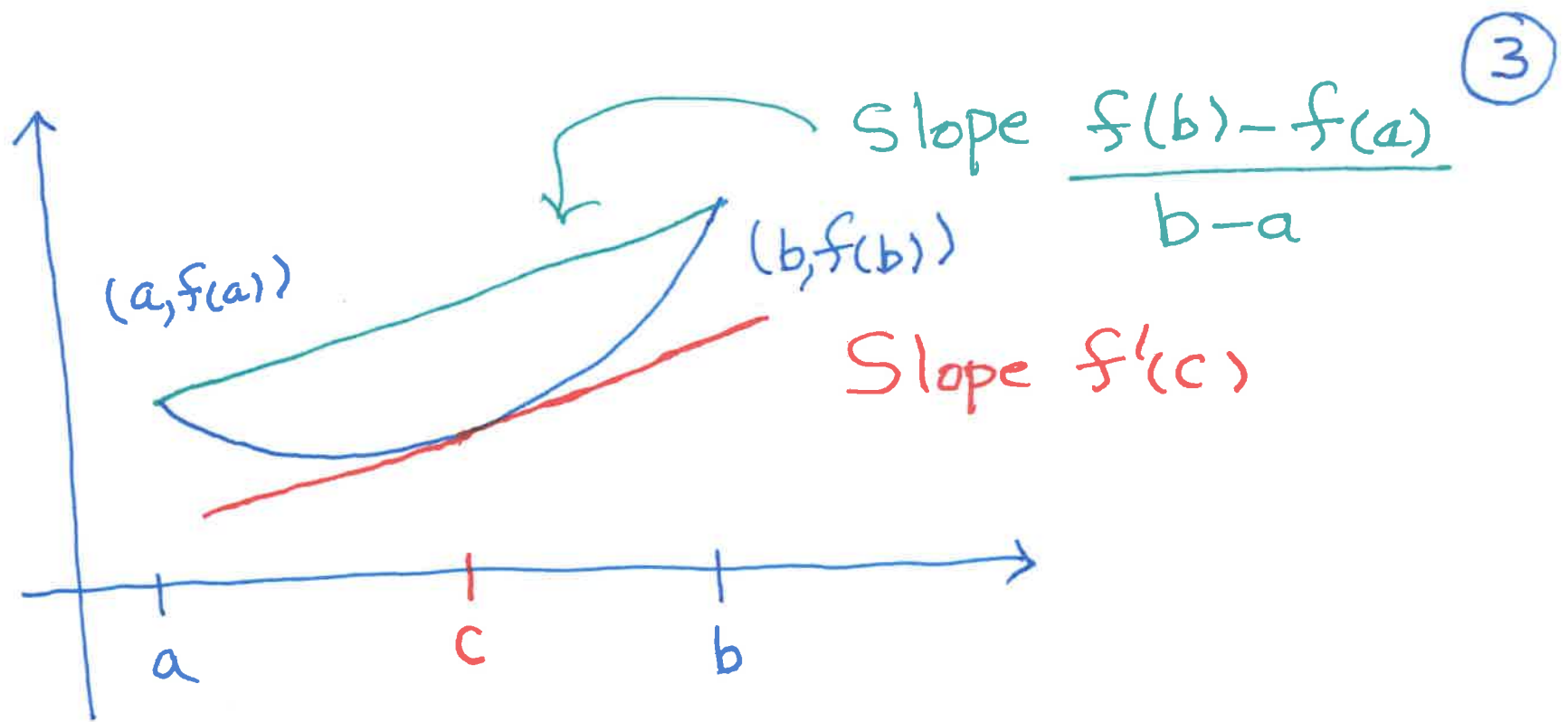
$$f(x) = f(x_0) + f'(c)(x - x_0)$$

$$x=b, x_0=a$$

$$f(b) - f(a) = f'(c)(b - a)$$

Mean Value Thm: Assume f is cont. on $[a, b]$ and diff on (a, b) . Then there exists $c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Can be more than
one c that works

④

Cor: Let I be an interval. Assume
 f is diff. on I .

a) If $f'(x) > 0$ on I then f is
increasing on I .

b) If $f'(x) < 0$ on I then f is
decreasing on I .

c) $f'(x) = 0$ for all $x \in I$ then
 f is constant on I .

⑤

Pf of (a): Assume $f'(x) > 0$ for all $x \in I$. Take $x_1, x_2 \in I$ with $x_1 < x_2$.

Need to show $f(x_1) < f(x_2)$.

Choose $c \in (x_1, x_2)$ (by MVT)

$$\text{s.t.} \quad f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

$$f'(c) > 0, \quad x_2 - x_1 > 0$$

$$f(x_2) - f(x_1) > 0 \quad \square$$

⑥

Max - Min

Let I be an interval. A function $f: I \rightarrow \mathbb{R}$ can attain a max (or a min) only at pts of the following types:

(i) Endpoints

(ii) Pts where $f' = 0$

(iii) Pts where f' DNE

⑦

Rmk: If I is a nonempty closed
bounded interval and f is cont.
on I then attains both a
max and a min on I .

⑧

2nd Deriv Test

Assume f is twice diff on a interval I . Let $x_0 \in I$ be given. Assume $f'(x_0) = 0$.

- (a) If $f''(x) \geq 0$ for all $x \in I$
then absolute min at x_0 .
- (b) If $f''(x) \leq 0$ for all $x \in I$
then absolute max at x_0 .

⑨

Proof of (a): Assume $f'(x_0) = 0$ and $f''(x) \geq 0$ for all $x \in I$.

Taylor's Theorem $n=1$

c depends
on x_0, x

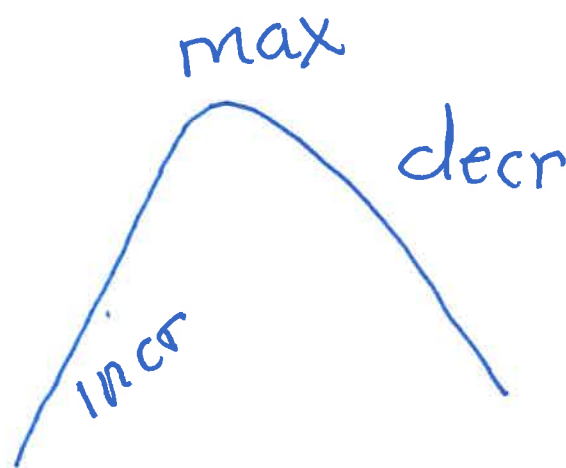
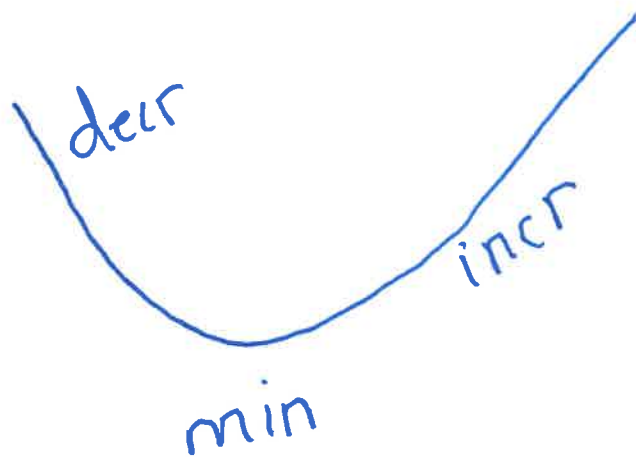
$$f(x) = f(x_0) + \cancel{f'(x_0)}(x-x_0) + \frac{1}{2}f''(c_{x_0, x})(x-x_0)^2$$

$$= f(x_0) + \underbrace{\frac{1}{2}f''(c_{x_0, x})}_{\geq 0} \underbrace{(x-x_0)^2}_{\geq 0}$$

$$f(x) \geq f(x_0) \quad \square$$

10

Can also use sign of f'
to locate max and min



11

Upper Level Mathematics

Concave Up $\longrightarrow$ Convex

Concave Down $\longrightarrow$ Concave

$x \longrightarrow \ln x$

Concave

$x \longrightarrow x^2$

Convex

(12)

Riemann Integration

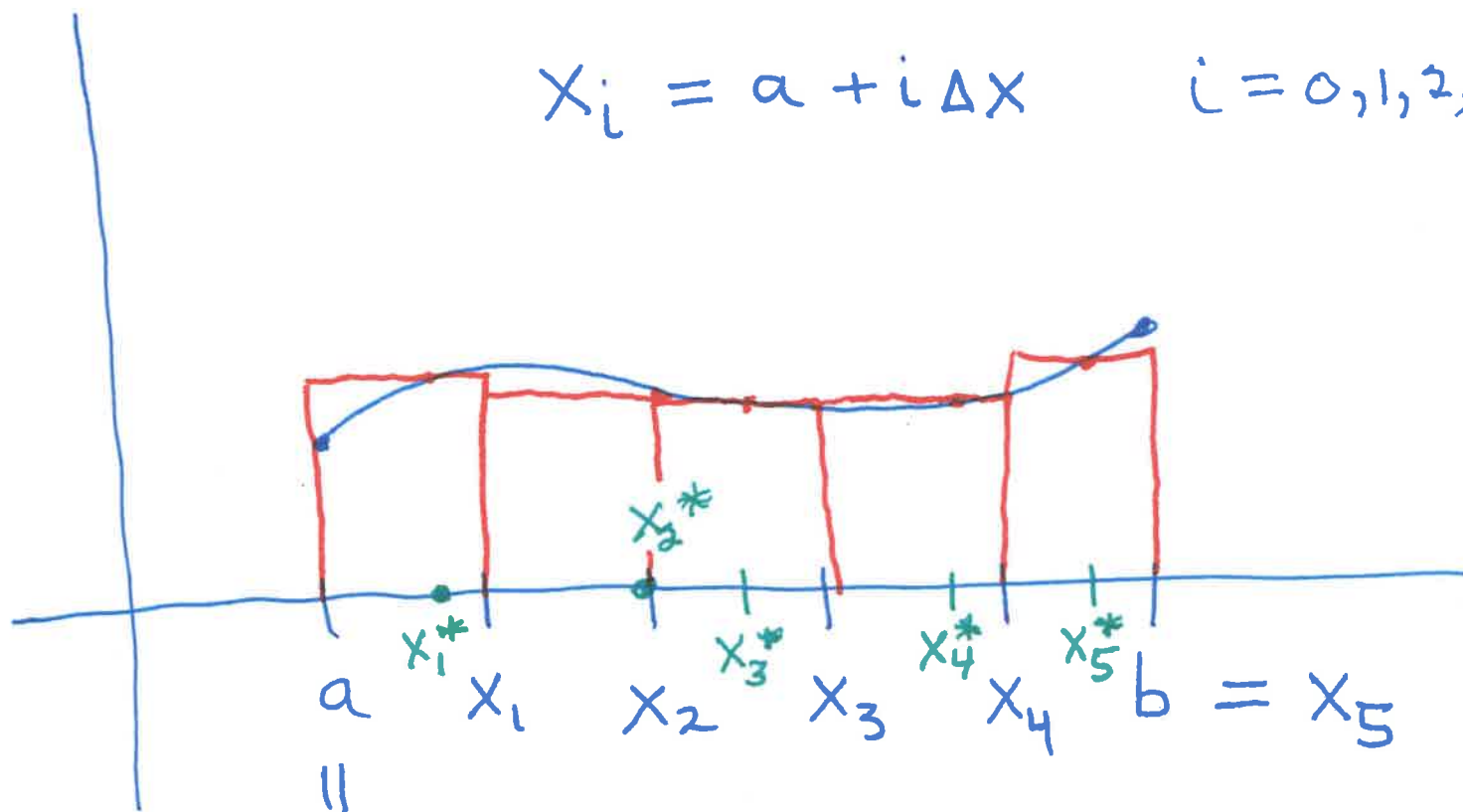
Assume $f: [a, b] \rightarrow \mathbb{R}$ is bounded and that f has at most a finite number of discontinuities.

Subdivide $[a, b]$ into n equal pieces, form a sum, then let $n \rightarrow \infty$.

(13)

$$\Delta x = \frac{b-a}{n} \quad x_0, x_1, \dots, x_n$$

$$x_i = a + i\Delta x \quad i = 0, 1, 2, \dots, n$$


 x_0

Choose $x_i^* \in [x_{i-1}, x_i]$

(14)

$$\sum_{i=1}^n f(x_i^*) \Delta x = \sum_{i=1}^n f(x_i^*) \left(\frac{b-a}{n} \right)$$

$$= \left(\frac{b-a}{n} \right) \sum_{i=1}^n f(x_i^*)$$

Riemann Sum

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \left(\frac{b-a}{n} \right) \sum_{i=1}^n f(x_i^*)$$

15

Can Choose x_i^* however I wish.

Common Choices

$$x_i^* = x_{i-1}$$

Left Endpoints

$$x_i^* = x_i$$

Right Endpoints

$$x_i^* = \frac{1}{2}(x_{i-1} + x_i)$$

Midpoints

(16)

REMARK :

For stochastic
integrals, how you choose x_i^*
matters a lot!!! For

Ito Integrals, use
LEFT ENDPOINTS.

Basic Properties

$$(i) \int_a^b c_1 f(x) + c_2 g(x) dx = c_1 \int_a^b f(x) dx + c_2 \int_a^b g(x) dx$$

$$(ii) \text{ If } c \in (a, b) \text{ then}$$
$$\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$$

$$(iii) \text{ If } f(x) \geq 0 \text{ for all } x \in [a, b]$$
$$\text{then } \int_a^b f(x) dx \geq 0 \quad \text{Need } a < b$$

$$[f(x) \geq g(x) \rightarrow \int_a^b f(x) dx \geq \int_a^b g(x) dx]$$

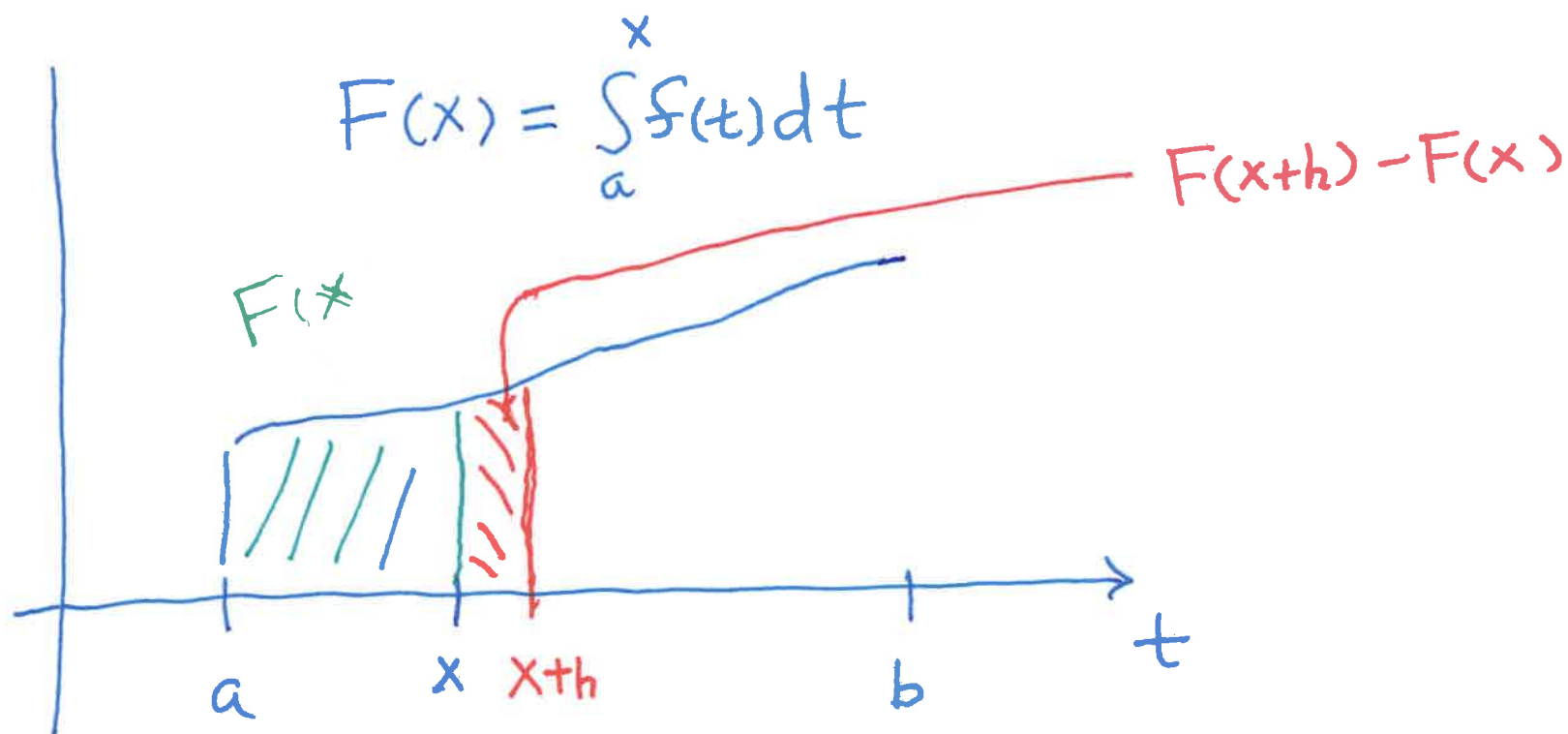
$$(iv) \int_a^b 1 \cdot dx = b - a$$

Fundamental Theorem of Calculus:

Assume F is differentiable on $[a, b]$
and $F'(x) = f(x)$ for all $x \in [a, b]$

$$\text{Then } \int_a^b f(x) dx = F(b) - F(a)$$

Rmk: $\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f$, etc.



(20)

Compute $F'(x)$

Look at $\frac{F(x+h) - F(x)}{h}$

$$\begin{aligned} F(x+h) &= \int_a^x f(t) dt + \int_x^{x+h} f(t) dt \\ &= F(x) + \int_x^{x+h} f(t) dt \end{aligned}$$

$$F(x+h) - F(x) = \int_x^{x+h} f(t) dt$$

(21)

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt$$

h is small $f(t) \approx f(x)$ on $[x, x+h]$

$$\frac{F(x+h) - F(x)}{h} \approx \frac{1}{h} f(x) \cdot h = f(x)$$

Let $h \rightarrow 0$ $F'(x) = f(x)$

Conventions

$$(i) \int_a^a f(x) dx = 0$$

$$(ii) \int_b^a f(x) dx = - \int_a^b f(x) dx$$

23

Differentiation of Integrals

Choose F s.t. $F' = f$
(Assume f is continuous)

Compute $\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt$

$$\int_{u(x)}^{v(x)} f(t) dt = F(v(x)) - F(u(x))$$

(24)

$$\begin{aligned} \frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt &= F'(v(x))v'(x) \\ &\quad - F'(u(x))u'(x) \\ &= f(v(x))v'(x) - f(u(x))u'(x) \end{aligned}$$

Rmk: $\frac{d}{dx} \int_a^x f(t) dt = f(x) \cdot \frac{d}{dx}(x) - \cancel{f(a) \frac{d}{dx}(a)} \rightarrow 0$

$$= f(x)$$

$$\frac{d}{dx} \int_x^b f(t) dt = -f(x)$$

(25)

Ex: $\frac{d}{dx} \int_{\sin x}^{x^3} e^{t^2} dt$

$$= e^{x^6} \cdot 3x^2 - e^{\sin^2 x} \cdot \cos x$$

Antiderivative or
Indefinite Integral

If $F'(x) = f(x)$ we write

$$\int f(x) dx = F(x) + C$$

$\nearrow$
constant of integration

Methods to Compute Indefinite Integrals

Substitution (Chain Rule)

Parts (Product Rule)

Basic Antiderivatives

$$a) \int x^p dx = \frac{x^{p+1}}{p+1} + C \quad p \neq -1$$

$$b) \int \frac{dx}{x} = \ln|x| + C$$

$$c) \int e^x dx = e^x + C$$

$$d) \int \cos x dx = \sin x + C$$

$$e) \int \sin x dx = -\cos x + C$$